

Special Relativity

Big Ideas

We have 4 vector objects k^μ which must satisfy $k^{\mu'} = \Lambda k^\mu$
 where Λ is the lorentz transformation satisfying
 $\hat{\Lambda}^{-1} \eta \hat{\Lambda} = \eta$

where $\eta = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

The lorentz transformation as a matrix represents a hyperbolic rotation (if the movement is only on two axis - 1 space and time), and can be written as a sum or a matrix in two equivalent ways:

$$\begin{aligned} X^{\nu'} &= \sum_{\mu=0}^3 \Lambda_{\mu}^{\nu'} X^{\mu} = \Lambda_{\mu}^{\nu'} X^{\mu} \\ &= \Lambda_0^{\nu'} X^0 + \Lambda_1^{\nu'} X^1 + \dots \end{aligned}$$

If we have a moving object with its personal frame Θ' and a lab frame which is static Θ , then we can transform between the coordinates

$$\underline{\underline{x}}_{x^{1'}}' = \gamma_v (x - vt) = \gamma_v \left(\underline{\underline{x}}_{x^1} - \frac{v}{c} \underline{\underline{ct}}_{x^0} \right) \quad (1)$$

$$y' = y \quad (2)$$

$$z' = z \quad (3)$$

$$ct' = \gamma_v \left(\underline{\underline{ct}}_{x^0} - \frac{v}{c} \underline{\underline{x}}_{x^1}' \right) \quad (4)$$

Lets define $\frac{v}{c} = \beta_v$
 as a matrix, we can write this as

$$\begin{pmatrix} x^{0'} \\ x^{1'} \\ x^{2'} \\ x^{3'} \end{pmatrix} = \begin{pmatrix} \gamma_v & -\beta\gamma_v & 0 & 0 \\ -\beta\gamma_v & \gamma_v & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} \quad (5)$$

As short hand, we just write

$$\begin{pmatrix} x^{0'} \\ x^{1'} \end{pmatrix} = \gamma_v \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \end{pmatrix} \quad (6)$$

This looks like the rotation matrix, but because we have two negative signs on the off diagonals its a hyperbolic rotation instead of a circular one. It looks like

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cosh \xi & \sinh \xi \\ \sinh \xi & \cosh \xi \end{pmatrix} \quad (7)$$

We have momentum 4-vector

$$P = m \underbrace{\vec{U}}_{\text{energy}} \quad (8)$$

$$U = \gamma_v mc^2$$

$$P^{\mu'} = \lambda_{\mu}^{\mu'} P^{\mu} \quad (9)$$

$$P^{\mu} = (P^0, P^1, P^2, P^3) \quad (10)$$

$$= \left(\underbrace{m\gamma_v c}_{\frac{U}{c}}, \underbrace{m\gamma_v \vec{V}}_{\vec{P}} \right) \quad (11)$$